Checking square, diagonals and layout

Day 21 • 2026-12-03 • 12:45 PM–2:00 PM

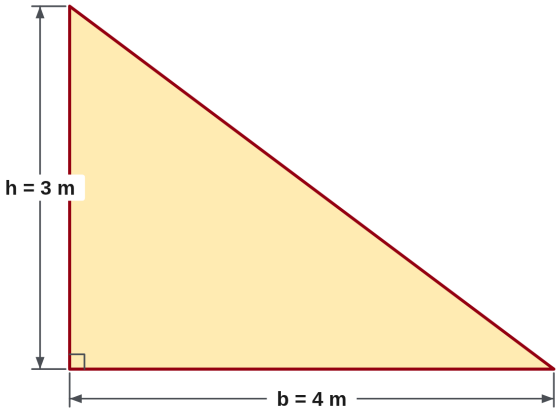
15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 READ THE LAYOUT

A 3–4–5 check uses the two legs shown in Shape Canvas. What diagonal makes the corner square?



Shape Canvas

Answer (m) _____ **Working / reason** _____

Textbook sections 12.5–12.6 pp.304–306

Checking square, diagonals and layout

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02 Which pair of lines should be measured when checking a rectangular frame?

A. Only the top edge and bottom edge

B. Only one diagonal

C. Only two short scraps

D. Both corner-to-opposite-corner diagonals

Choice (A–D)

Working / reason

Textbook sections 12.5–12.6 pp.304–306

Checking square, diagonals and layout

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03 When using a Numeracy Hub measurement diagram, which reading should be recorded?

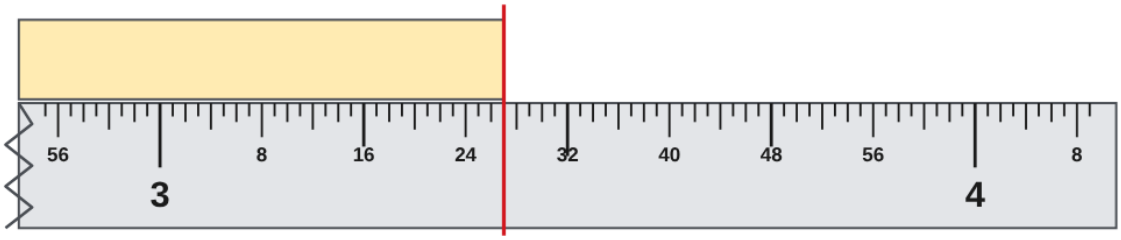
A. The number of pixels

B. A guessed number from the picture size

C. The stated or measured dimension with its unit

D. The screen length in centimetres

The part starts at the end of the rule (0).



Numeracy Hub

Choice (A–D)

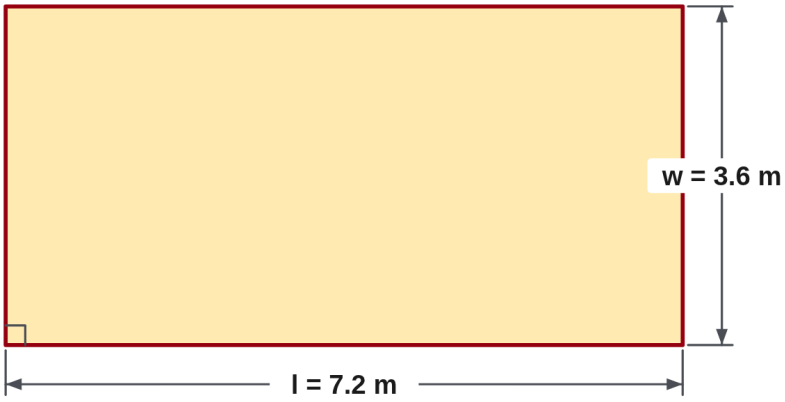
Working / reason

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04 The floor in Shape Canvas is 7.2 m by 3.6 m. What diagonal length is expected for a rectangle? Round to 2 decimal places.



Shape Canvas

Answer (m)

Working / reason

Textbook sections 12.5–12.6 pp.304–306

05 Which set is a scaled 3–4–5 check?

A. 900 mm, 1200 mm, 1500 mm

B. 900 mm, 1200 mm, 1800 mm

C. 300 mm, 400 mm, 600 mm

D. 600 mm, 900 mm, 1200 mm

Choice (A–D)

Working / reason

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06 TARGET DIMENSIONS

A scaled 3–4–5 check has a shorter leg of 1.8 m. Find the corresponding longer leg.

Answer (m) | **Working / reason**

Textbook sections 12.5–12.6 pp.304–306

07 A 3–4–5 square check has a 1.8 m shorter leg. Find its diagonal target.

Answer (m) | **Working / reason**

Textbook sections 12.5–12.6 pp.304–306

08 Two frame diagonals measure 2508 mm and 2496 mm. How much do they differ?

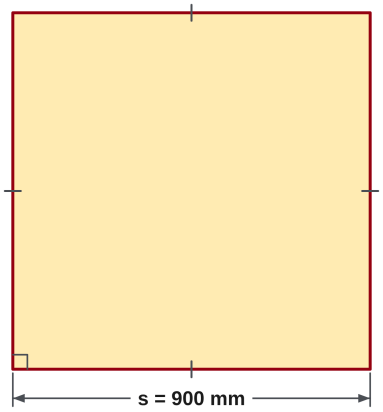
Answer (mm) | **Working / reason**

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09 A 900 mm square is shown. Find its theoretical diagonal. Round to 1 decimal place.



Shape Canvas

Answer (mm)

Working / reason

Textbook sections 12.5–12.6 pp.304–306

10 A rectangular opening has sides of 1.5 m and 2 m. Find the target diagonal.

Answer (m)

Working / reason

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11 CHECK FOR SQUARE

For a fixed two-leg layout, the distance between the free endpoints exceeds the right-angle target. What does that indicate?

A. The included corner is more open than 90 degrees

B. The included corner is less than 90 degrees

C. The target must be ignored

D. The legs have automatically shortened

Choice (A–D)

Working / reason

Textbook sections 12.5–12.6 pp.304–306

12 With two leg lengths held fixed, their endpoint distance is below the right-angle target. How should the corner change?

A. Close the corner further

B. Shorten both legs without checking

C. Change the units only

D. Open the included corner toward 90 degrees

Choice (A–D)

Working / reason

Textbook sections 12.5–12.6 pp.304–306

13 A 3 m by 4 m layout shows 5.03 m across the endpoints. How far is this above the square target?

Answer (mm)

Working / reason

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14 The two measured diagonals differ. What is a useful next action?

A. Hide the larger reading

B. Recheck the measurements and adjust the layout before repeating both diagonals

C. Cut every piece immediately

D. Average the readings and declare the frame square

Choice (A–D)

Working / reason

Textbook sections 12.5–12.6 pp.304–306

15 Which record gives the clearest evidence for a square check?

A. Side lengths, target diagonal, both measured diagonals and units

B. A photograph without dimensions

C. Only the final word square

D. A single calculator display with no diagram

Choice (A–D)

Working / reason

Textbook sections 12.5–12.6 pp.304–306